ENSAE NLP Project - Topic Modeling on Archelec Corpus

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Abstract

In this project, we apply topic modeling techniques to the Archelec corpus, a collection of political manifestos from candidates in the French 'élections législatives' between 1973 and 1993. We compare three approaches: Latent Dirichlet Allocation (LDA), Non-negative Matrix Factorization (NMF), and BERTopic. We show that all three methods successfully recover politically meaningful themes, several of which can be directly linked to the discourse of specific political parties. We also discuss the limitations of each approach and identify directions for improvement, including more robust preprocessing.

1 Introduction

The Archelec Corpus [1] is a collection of political manifestos of candidates in the elections in the French Vth Republic. In this project, we focus on the "élections législatives" between 1973 and 1993.

The objective of this project is to analyze these documents using topic modeling techniques. Topic modeling is a family of tools used to automatically emphasize hidden themes, called topics, in a corpus of documents. Such methods are particularly well-suited to this corpus, as political manifestos tend to mention recurring thematic clusters such as social policy, economy or national identity.

We compare three topic modeling approaches: Latent Dirichlet Allocation, Non-negative Matrix Factorization and BERTopic. The remainder of this report is organized as follows: Section 2 presents the methods, Section 3 describes the data and preprocessing steps, Section 4 reports and discuss the results of the three methods and Section 5 concludes.

2 Methods

In this section, we will present the main topic modeling methods used throughout this project.

2.1 A classical probabilistic model - Latent Dirichlet Allocation

Latent Dirichlet Allocation (LDA) is a Bayesian probabilistic model of text document, introduced in [1]. Each document is modeled as a mixture of topics which are also modeled as a mixture of topic probabilities over words.

More formally, the underlying generative process is the following. For each document d, draw $\theta_d \sim \mathrm{Dirichlet}(\alpha)$. Then, for each word i in this document d, begin by drawing a topic $z_{d,i}$ from a multinomial distribution with parameter θ_d and then a word $w_{d,i}$ from a multinomial probability with parameter β conditioned on the topic $z_{d,i}$. This generative process is described in Figure 1.

A corpus of documents is analyzed through this probabilistic model by examining the posterior distribution of the topics and topic proportions. This posterior is not tractable and is usually approximated

[1] https://archelec.sciencespo.fr/

40th Conference on Neural Information Processing Systems (NeurIPS 2026).

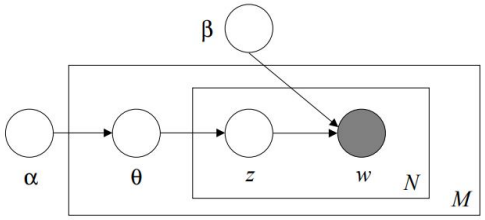


Figure 1: Probalistic Graphical Model representation of LDA. Image coming from [1, Figure 1]. M is the number of documents and N is the number of word per document.

using variational inference. In this project, we use the implementation of the scikit-learn [2] package based on [4, 5].

This model assumes a bag-of-words representation and only rely on word frequencies, obtained with vectorization methods.

2.2 Non-negative matrix factorization

Non-negative matrix factorization (NMF) is a linear algebra-based topic modeling technique that decomposes a document-term matrix into two lower-dimensional non-negative matrices. It was originally proposed in [6] as a representation learning method and has been shown to extract semantic features from text and thus enabling a meaningful topic representation of a corpus.

As the LDA method presented previously, this method relies also only on word frequencies, without using the position of words. Indeed, NMF uses document-term matrix, obtained with TF-IDF (Term Frequency-Inverse Document Frequency) which scores each word in each document, indicating how important a word is to a document, relative to the corpus it belongs to.

Mathematically, we denote by $X \in \mathbb{R}^{M \times n_f}$ this document-term matrix, where M is the number of documents and n_f is the number of features. Denoting by n_t the number of topics, we seek $W \in \mathbb{R}^{M \times n_t}$ and $H \in \mathbb{R}^{n_t \times n_f}$, both non negatives, such that WH is the "closest" approximation to X. Here, W represents the importance of each topic in each document, while H represents the importance of each word in each topic.

The topics extracted from NMF are easily interpretable using the H matrix which describes the words associated with each topic. The matrix W can then be used to analyze the presence of topics in each document.

2.3 An embedding-based method - BERTopic

The main drawback of the previously presented methods is that they only rely on the frequencies of the words and not on their positions. To address this issue, BERTopic is a topic modeling approach based on contextual embeddings. This enable to use the position of the words and therefore identify semantic similarities between documents of the corpus.

It has been introduced in [3]. The texts are first embedded with a pre-trained transformer-based language model. For this task, we use the pre-trained model Sentence-BERT [11], available on HuggingFace [3]. Then, the embeddings are clustered using HDBSCAN [7] (Hierarchical Density-Based Spatial Clustering of Applications with Noise). Finally, BERTopic uses a vectorizer to identify the different topic labels.

[2] https://scikit-learn.org/stable/modules/generated/sklearn.decomposition.LatentDirichletAllocation.html
[3] https://huggingface.co/sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2

2.4 Coherence metrics to select the number of topics

In all the previously mentioned methods, one crucial parameter to take into account is the number of topics considered. Selecting an appropriate number of topic is crucial because it affects the interpretability of the discovered themes. Too few topics may merge not semantically related concepts into a single topic, while too many may create unnecessary division within what could be coherent topics. To address this issue, several metrics have been designed, in order to evaluate the quality of topics, referred to as coherence metrics.

In our implementation, we use the gensim library [10] and the four provided following metrics:

- `u_mass` [8], which measures word co-occurrence based on document frequencies within the corpus;
- `c_v` [12], which combines sliding-window co-occurrence and normalized pointwise mutual information (NPMI) with a cosine similarity measure;
- `c_uci` [9], which relies on pointwise mutual information (PMI) computed over word co-occurrences;
- `c_npmi` [2], which is a normalized version of PMI that scales scores between -1 and 1 to improve interpretability.

3 Data

The data consists of the political manifestos of candidates for the french "élections législatives" of 1973, 1978, 1981, 1988 and 1993. They have been OCRed (Optic Character Recognition) and are available in the repository [4].

3.1 Statistical study of the data

First, we conducted a statistical study of the data, using the metadata file in CSV format. This file indicates a total number of around 21k text files. In the Figure 2, we have represented the distribution of this number of text files per year. For each of these years, we have between 3000 and 6000 text files.

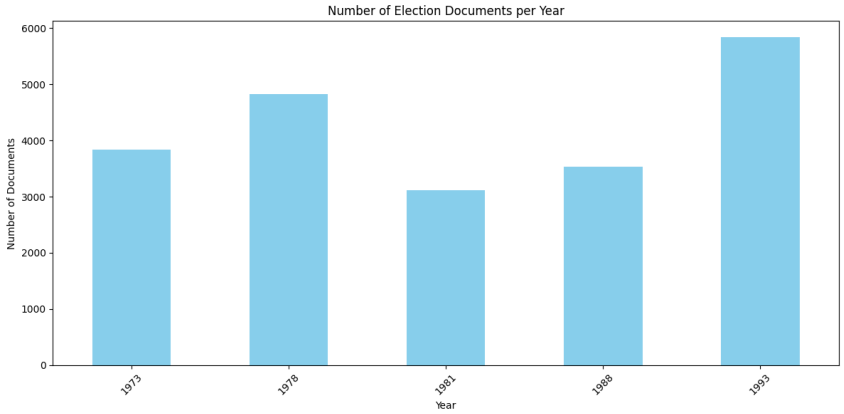


Figure 2: Number of text files per year

For the rest of the study, one variable which can be of the interest is the possible political party affiliation of the candidates. In Figure 10 in the appendix, we have represented the number of candidates for each political party in the top 15. The first party here 'non mentionné' is not actually a party but deals with the fact that roughly 20% of the candidates are not affiliated to a party (or don't mention it). We have done the same type analysis specifically in the year 1973 in Figure 11 for the sake of comparison, which shows that in 1973, candidates tend to mention more their affiliation to a political party.

[4] https://gitlab.teklia.com/ckermorvant/arkindex_archelec

3.2 Text files and preprocessing

We can now turn to the analysis of the text files itself. We extract the text and begin by preprocessing removing stopwords. After this removal of stopwords, we plot in Figure 3 the most frequent words in the corpus. We see that there remains words that are non-informative for the task of topic modeling.

First, 'Sciences', 'Po', 'fonds' and 'CEVIPOF' are just words added by the OCR task by SciencesPo and need to be removed like stopwords.

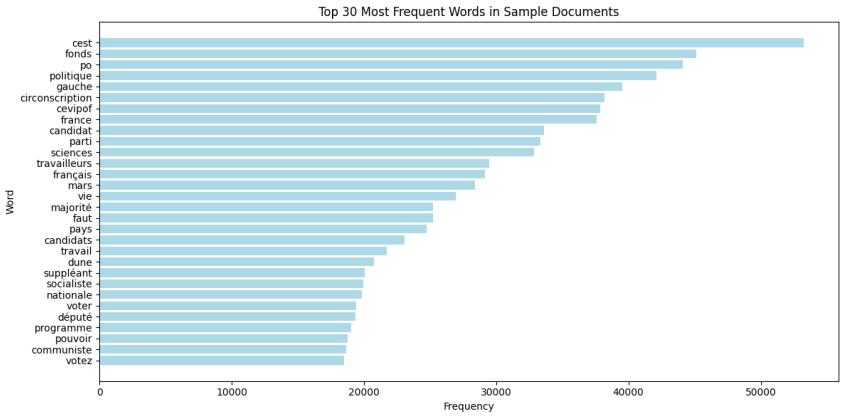


Figure 3: Most frequent words in the corpus after having removed stopwords.

3.3 Selecting the number of topics

For each coherence metric, we calculate the coherence score for various values of k, using the NMF method presented in Section 2.2. Since these metrics have different scales and are not directly comparable, we apply min–max normalization independently to each metric. This enables us to compare their relative trends across different numbers of topics.

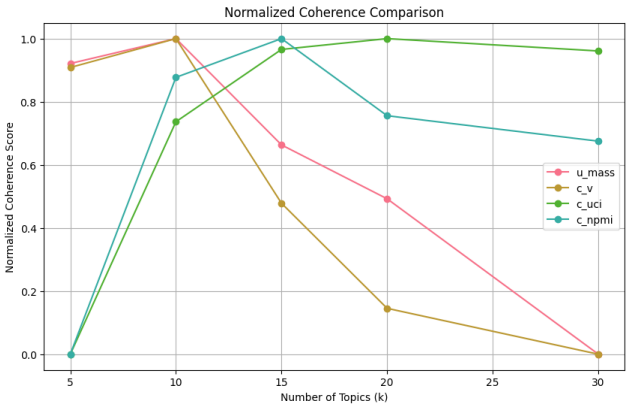


Figure 4: Coherence metrics Depending on the chosen number of topics.

In Figure 4, we have represented the coherence scores as a function of the number of topics. For each of these metrics, the higher, the better. Therefore, we see that the optimal number of topics seems to be between 10 and 15. For the following of the study, we will fix the number of topics as 10 in order to compare the three proposed methods of topic modeling.

4 Results

4.1 LDA

As mentioned before, we have chosen a number of topics of 10 to conduct LDA on our corpus. After fitting the model provided by scikit-learn, we obtain these topics. For each of the topic, we can look at the most representative words of this topic, in order to interpret the semantic behind the topic. The topics obtained are showed in Table 1.

We can clearly extract topics which are related to some political party. For example, the topic number 5 is clearly related to the 'Parti Communiste Français' with the words 'lutte', 'travailleur', 'ouvrière' and 'marchais' for George Marchais which was the head of this party in the 70s and 80s. We can see the same pattern with the topic 8 which seems to be related with the 'Front National' mentioning 'immigration' and 'insécurité'.

Topic	10 most representative words
0	die, der, und, für, in, zu, das, den, eine, sie
1	vie, pouvoir, écologiste, écologie, droit, politique, travail, vouloir, femme, ps
2	candidat, circonscription, majorité, france, député, juin, maire, socialiste, union, conseiller
3	rpr, udf, français, voter, impôt, nature, aller, million, 1993, mars
4	travailleur, pouvoir, falloir, entreprise, devoir, payer, patron, ouvrier, emploi, lutte
5	circonscription, suppléant, marchais, lutte, candidat, travailleur, ouvrière, travailleuses, luire, voter
6	social, politique, devoir, pouvoir, france, emploi, vie, pays, national, économique
7	mitterrand, voter, travailleur, gauche, petit, femme, parti, pouvoir, candidat, gouvernement
8	national, français, front, immigration, france, immigré, vote, famille, voix, insécurité
9	communiste, parti, gauche, politique, pouvoir, programme, voter, commun, candidat, changement

Table 1: 10 most representative words for each topic given by the LDA model.

These results seem quite promising but on the other hand, some topics does not seem to be really informative. The topic 0 seems to be containing german words, which is kind of weird. The topic 2 contains basic words related to the 'élections législatives' like 'candidat' or 'circonscription'. These words could have maybe been added to the list of stopwords since they are non-informative in the context of our corpus. However, this is not the choice that has been made here.

Besides, some topics seem related to each other. For example the topics 4, 5, 8 and 9 are all related more or less to the vocabulary of social policies which are traditionally linked to demands of the left-wing parties. In order to visualize more this aspect, we can use the pyLDAvis library [13]. This is confirmed in the Figure 5, where these topics are the number 3, 5, 8 and 9, which are close to each other in the proposed representation space.

4.2 NMF

We can now do the same analysis with the NMF topic modeling method. As before, we use 10 topics. Like for the LDA model, we also find a topic corresponding to german words. Apart from this topic, we can now give a name to the different topic and using the matrix W given by the NMF, we can analyze the distribution of topics for each of the document of the corpus.

An example of such a text and the associated distribution of topics is given in Figure 6 and Figure 7. In this example, the distribution of topics seem to match well the text, since the most dominant topic is 'Communiste' and the candidate writing this manifesto is affiliated to the communist party. However, this matching is not always perfect as demonstrated in a second example, put in the Section B of the appendix.

Overall, we can analyze these results by crossing the results with the political party given in the metadata file. For this task, we will extract the lines corresponding to a specific french political party and look at the corresponding distribution of dominant topic associated to each of the text files. An example of such an analysis is shown on Figure 8 with the example of the Front National.

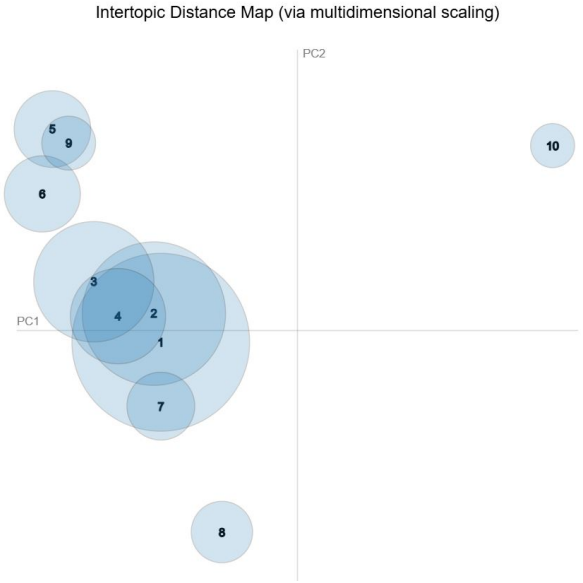
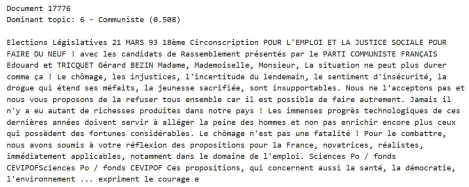


Figure 5: Screenshot of the pyLDAvis visualization of the 10 topics. The numbers associated to each topics are not the same as the order of Table 1.



Document 17776
Dominant topic: 6 - Communiste (0.508)

Elections Législatives 21 MARS 93 18ème Circonscription POUR L'EMPLOI ET LA JUSTICE SOCIALE POUR FAIRE DU NEUF ! avec les candidats de Rassemblement présentés par le PARTI COMMUNISTE FRANÇAIS Edouard et TRICQUET Gérard BEZIN Madame, Mademoiselle, Monsieur, La situation ne peut plus durer comme ça ! Le chômage, les injustices, l'incertitude du lendemain, le sentiment d'insécurité, la drogue qui étend ses méfaits, la jeunesse sacrifiée, sont insupportables. Nous ne l'acceptons pas et nous vous proposons de la refuser tous ensemble car il est possible de faire autrement. Jamais il n'y a eu autant de richesses produites dans notre pays ! Les immenses progrès technologiques de ces dernières années doivent servir à alléger la peine des hommes et non pas enrichir encore plus ceux qui possèdent des fortunes considérables. Le chômage n'est pas une fatalité ! Pour le combattre, nous avons soumis à votre réflexion des propositions pour la France, novatrices, réalistes, immédiatement applicables, notamment dans le domaine de l'emploi. Sciences Po / fonds CEVIPOPSciences Po / fonds CEVIPOP Ces propositions, qui concernent aussi la santé, la démocratie, l'environnement ... expriment le courage e

Figure 6: An example of a text of the corpus.

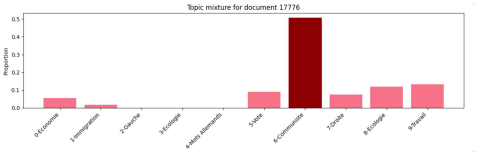


Figure 7: The corresponding topic distribution.

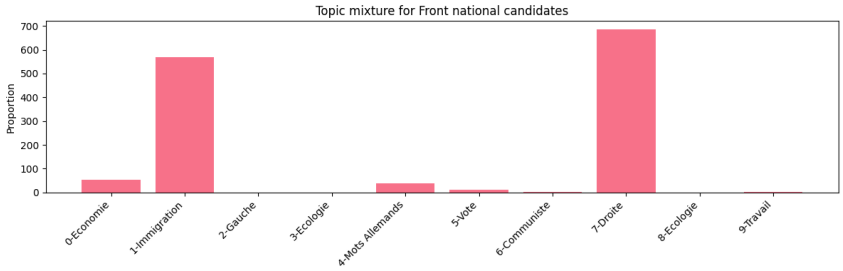


Figure 8: Repartition of the most dominant topic given by the NMF for the candidates of Front National.

We see that overall the extracted dominant topic corresponds well to the affiliation of the candidate with the Front National candidates being associated to the topic 'Droite' or the topic 'Immigration'. Similar conclusions can be drawn from other parties as can be seen in the Figures 14 and 15 in the appendix.

4.3 BERTopic

Finally, we applied the BERTopic strategy to our corpus. The visualization of the result is in Figure 9. As before, we find topics related to themes treated by specific political parties. However, the topic 0, with representative words 'gauche', 'politique' and 'france' seem very generic and contains a huge proportion of the documents. This topic is thus not very instructive and takes a prominent place in the visualization, which makes difficult to conduct a more thorough analysis.

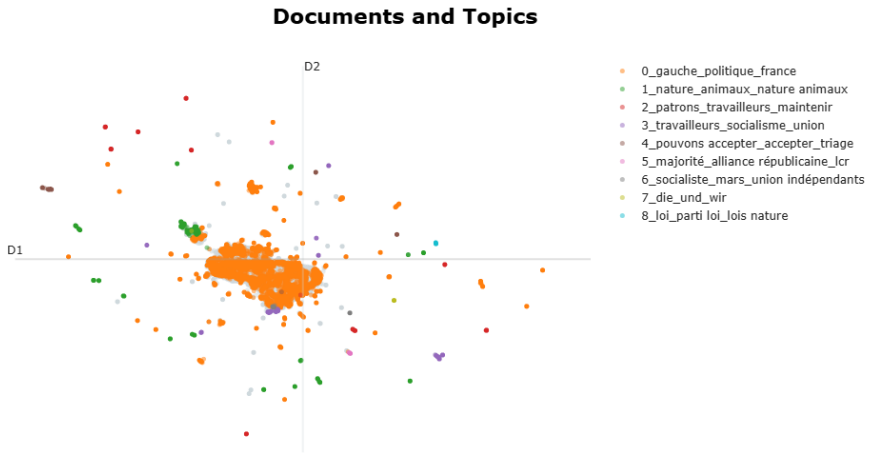


Figure 9: BERTopic visualization of the 10 extracted topics.

5 Conclusion

In this project, we applied three topic modeling methods to the Archelec corpus of French legislative elections manifestos from 1973 to 1993. All three methods successfully identified politically meaningful topics. Indeed, they recovered themes closely associated with specific parties. The NMF method seems to give particularly interpretable results, as its topic-document matrix allows for a direct analysis of topic distribution per candidate and per party. Linking this the metadata confirmed the fact the extracted topics align well with the political affiliation of the candidates.

However, there are several limitations to this work. First, all three methods produced a topic dominated by German words, suggesting that some texts should have been filtered out of the corpus to ease the analysis. Second, some topics capture generic electoral vocabulary rather than informative political content. This is particularly the case with the BERTopic model where a largely dominant topic absorbed a large share of the documents and limits the precision of the analysis. To address this issue, it could certainly be useful to extend the list of stopwords with this type of generic terms.

References

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A Documents per party

A.1 Overall years

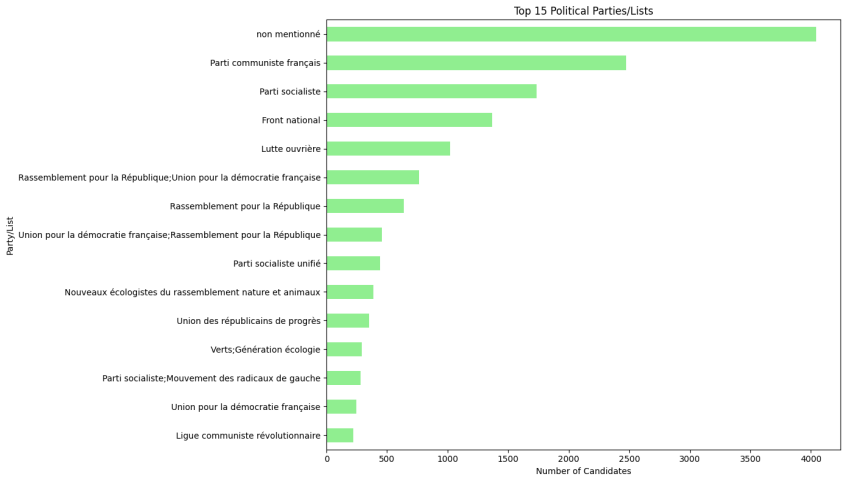

Figure 10: Number of text files per political party.

A.2 In 1973

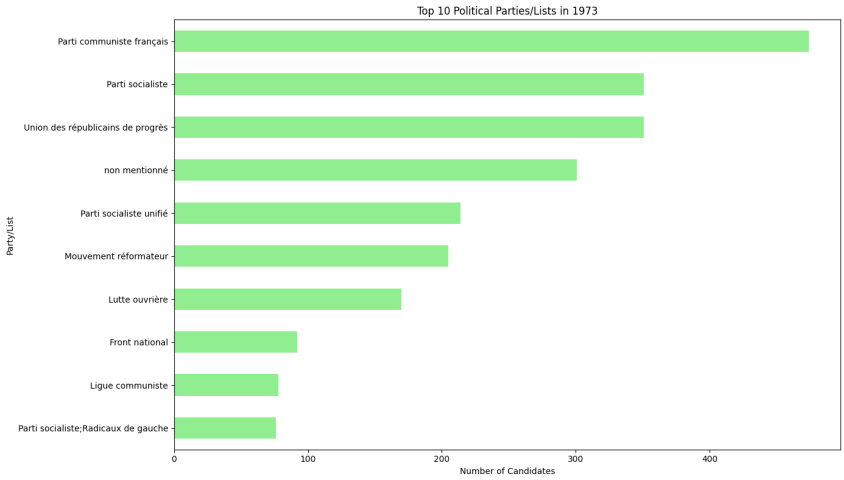

Figure 11: Number of text files per political party in 1973.

B NMF bad matching

Document 2127
Dominant topic: 6 - Communiste (0.504)

ELECTIONS LEGISLATIVES DU 4 MARS 1973 23ª circonscription de Paris - 17e arrondissement (Plaine Monceau-Batignolles) CENTRE ET RÉFORMATEURS - CENTRE DÉMOCRATE (Jean LECANUET) - MOUVEMENT RÉFORMATEUR - CENTRE NATIONAL DES INDÉPENDANTS (Président d'honneur Antoine PINAY) Soutien PRESENTENT : PAUL GARSON Avocat à la Cour Vice-Président du Conseil Municipal de Paris (1961-1962) Président de la Commission des Anciens Combattants du Conseil Général (1959-1965) Croix de Guerre (1939-1945) Campagnes d'Afrique, de France et d'Allemagne Ancien de la 1re Armée Française (Rhin et Danube) SUPPLEANT : Yves de DARUVAR Cadre Administratif Grand Officier de la Légion d'honneur Compagnon de la Libération Ancien de la Division Leclerc LE CHANGEMENT DANS LE CALME Les Français aspirent au changement, mais ils veulent, dans leur grande majorité, ainsi que l'a déclaré Jean LECANUET, qu'il s'opère dans le calme, sans bouleversement, sans crise politique, sans remise en cause des institutions qu'ils se sont données. LE CHOIX ENTRE TROIS COURANTS Le prochain scrutin pour le renouvellement de l'Assemblée Nationale leur présente un choix simple entre trois courants principaux : l'U.D.R. qui détient le pouvoir

Figure 12: An example of a text of the corpus.

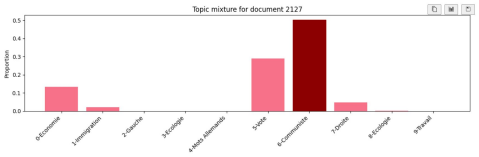

Figure 13: The corresponding topic distribution.

C Topic mixtures for candidates of specific political parties

C.1 Candidates of Lutte Ouvrière

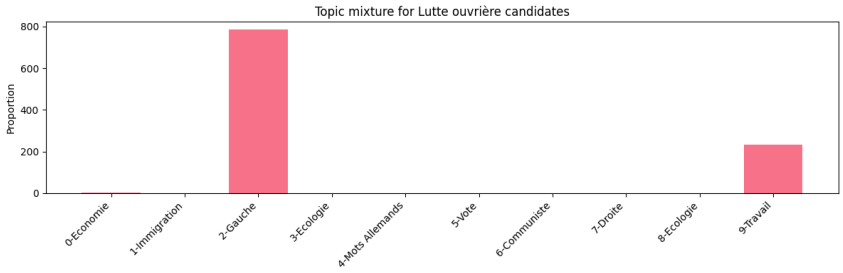

Figure 14: Repartition of the most dominant topic given by the NMF for the candidates of Lutte Ouvrière.

C.2 Candidates of Verts/Génération écologie

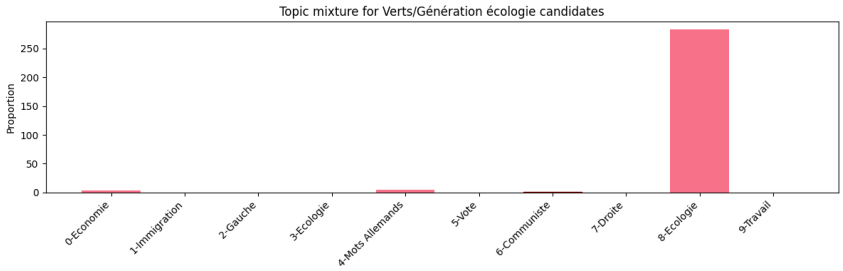

Figure 15: Repartition of the most dominant topic given by the NMF for the candidates of les Verts/Génération écologie.